

Maksim Shmalts

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I am pursuing a PhD in computational linguistics at the University of Tübingen. My primary interests include weak constraints in grammar, abstraction/generalization capabilities of LLMs, and pragmatic reasoning with LLMs. In particular, I am currently investigating whether LLMs can abstract the interaction of conflicting constraints that govern irregular phenomena in German as humans do.

Related Work Experience

- 2025- **Research Assistant**, Eberhards Karls Universität Tübingen, Tübingen, on-site
- 2029 • Currently occupied at a long-term project called The Status of Linguistic Constraints in Neural Language Models as a part of the [LaSTing Priority Program](#) (funded by the DFG, reference number HI 495/8-1). The current step of the project is investigating whether the LLMs are able to acquire weak (defeasible) grammar constraints on the example of German compounding. See [DieKonstraints](#) on GitHub.
- Instructed the course "LLM-based Assistants" for BA students. See the [course page](#) and the [repo](#) on GitHub.
- 2024- **Working Student for LLMs**, IPH gGmbH, Hannover, remote
- 2025 • Prototyped custom RAG pipelines (Python, n8n, LangChain, streamlit, docker).
- 2023- **Working Student for LLMs**, IAV GmbH, Berlin, hybrid
- 2024 • Implemented and integrated an internal tool for evaluation of LLMs and RAG pipelines (Python, LangChain, Ragas, OpenAI).
- Connected OpenAI tool calling to our internal on-premise models (Python, TGI, LangChain).
- 2021- **Data Scientist / NLP Engineer**, MKSKOM, Moscow, hybrid
- 2023 • Implemented the backend for a custom RAG engine for a large customer (Python, LangChain, Llama.cpp, HF Transformers).
- Increased the number of responses from potential customers on a freelance platform by 5 times by designing and implementing an LLM-based tool for automatic search for and contacting leads (Python, LangChain, OpenAI, Puzzle [internal tool]). Communicated with potential customers on that platform.
- Implemented prototypes of various tools for NLP, time series prediction, optimization modeling, data analysis (Python, PyTorch, spaCy, scikit-learn, statsmodels, Pandas, NumPy, streamlit).
- 2022- **Tutor**, Eberhards Karls Universität Tübingen, Tübingen, on-site
- 2023 • Held tutorials and contributed into evaluating assignments and projects for the courses "String Algorithms" (WiSe22/23), "Python for Beginners" (WiSe22/23), "Statistical Language Processing II" (SoSe23), "Reinforcement Learning for LLMs" (SoSe24).
- Created Finite-States Transducers for measuring distance between German dialects for the course "String Algorithms" (WiSe22/23) (Python, pynini).

Publications

- 2026 **Constraints on Linking Element Choice in German Nominal Compounding: A Large-Scale Corpus Study**
In Proceedings of the Workshop on Structured Linguistic Data and Evaluation (SLiDE), DOI: [10.63317/24uhjb79y8ov](https://doi.org/10.63317/24uhjb79y8ov).
- 2024 **Derbi: Deutscher regelbasierter Inflektor**
In [Studia Slavica XX](#), Tallinn, ISBN 2674-2160.
- 2023 **John Boy Walton at SemEval-2023 Task 5: An Ensemble Approach to Spoiler Classification and Retrieval for Clickbait Spoiling**
In Proceedings of the 17th International Workshop on Semantic Evaluation (SemEval-2023), Toronto, DOI [10.18653/v1/2023.semeval-1.289](https://doi.org/10.18653/v1/2023.semeval-1.289).

Education

- 2025- **Eberhard Karls Universität Tübingen**, PhD: Computational Linguistics, Tübingen
- curr • 2026 Talk and [publication](#) at the [SLiDE workshop](#) at the [LREC26](#) conference.
- 2025 Was admitted to the PhD program directly after completing BA.
- 2022- **Eberhard Karls Universität Tübingen**, BA: Computational Linguistics, Tübingen
- 2025 • 2024 Bachelor thesis on applying LLMs to German compound splitting (Python, LangChain, PyTorch, HF transformers). See [DEKOR](#) on GitHub.
- 2023 Participated at the SemEval 2023 and published a [result paper](#) (Python, HF transformers).
- 2019- **Lomonosov Moscow State University**, BA, interrupted: Theoretical and Applied Linguistics, Moscow
- 2022 • 2022 Developed DERBI: a tool for automatic inflection of German words (Python), see [DERBI](#) on GitHub.
- 2022 Participated at student conferences in Tallinn (on-site), Moscow (remote), Kaliningrad (remote).
- 2021 Term paper on predicting ablaut classes of German strong verbs (Python, PyTorch).

2017- **Lomonosov Moscow State University Gymnasium**, High School: History and Philology, Moscow
2019 • 2019 Prize-winner of All-Russian Olympiad for Linguistics (allows an exam-free admission to a linguistics program at the university of choice).
• 2019 Graduation with a Gold Medal.

Misc

Natural languages	Russian (native), English (C1), German (B2/C1), Spanish (B1), Italian (A2).
Programming languages	Python, R, C#, Java.
Tech stack	LangChain, HF Transformers, spaCy, PyTorch, scikit-learn, Pandas, NumPy; Docker, GitLab (CI/CD), basic Unix tools.
Misc achievements	Prize winner for numerous national and international musical competitions, driving license categories A and B, startup founder (few unsuccessful attempts).
Interests	Composing music, traveling, arts & crafts, table tennis, gardening, history, philosophy.